AQUORA
Community Resilience for Water & Climate

MANOUBA SCHOOL OF ENGINEERING

AQUORA

A living lab & geospatial platform for water resilience, food security, and climate adaptation across Africa

From satellite data to community action in one platform

Abundant rivers, scarce water security.



AFRICA

9%

OSS / UN Water (2023)

of the world's renewable freshwater resources while hosting 17% of the world's population.

TUNISIA

480 m³

World Bank (2023)

renewable water availability per person per year — the lowest in the MENA region.

CÔTE D'IVOIRE

1,160 m³

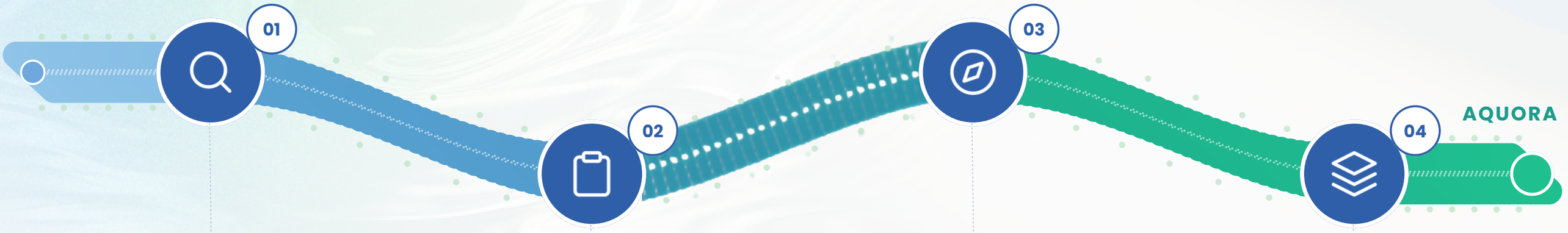
World Bank (2023)

renewable water availability per person per year — below the water stress threshold.

What if businesses, citizens, scientists, and governments could see – and act on – water risk together, in one place?

The Path to AQUORA

Follow the river — from scattered needs to one living platform



Analyzing User Needs

Studying each user group closely citizens, farmers, researchers, businesses to surface real, day-to-day water challenges.

Use Case Conception

Translating needs into concrete use cases, mapping how each role interacts with the platform.

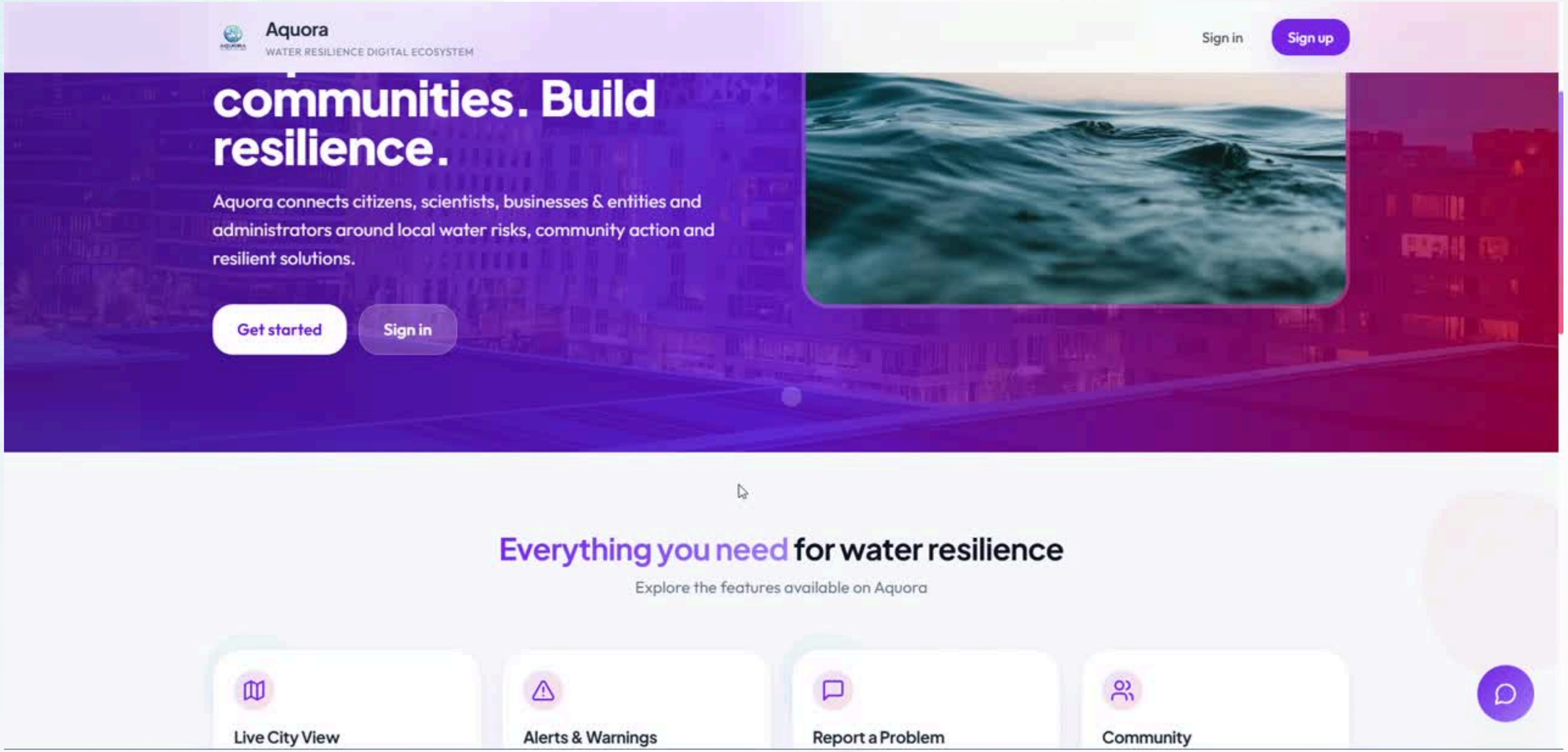
Living Lab Inspiration

Drawing from the Living Lab concept co-creating and testing with real users in real environments.

One Integrated Platform

Bringing every use case together into one comprehensive, living platform.

Introducing AQUORA



Data Sources

Every layer of Aquora is grounded in real, verifiable data — live feeds, open science, and reproducible models

Live Weather & Climate

- Open-Meteo — real-time forecast (14-day)
- ERA5 historical archive — 10 yrs daily rainfall
- CHIRPS v2.0 (UCSB) — configured, upcoming

Terrain, Satellite & Maps

- SRTM + ASTER DEM — 30m elevation tiles
- NASA EOSDIS GIBS — MODIS, VIIRS imagery
- MS Planetary Computer — Sentinel-2, Landsat
- OpenStreetMap, Nominatim, OpenFreeMap, Carto

Scientific Models

- SPI drought index — gamma-fit, McKee scale
- ML rainfall forecast — gradient boosting
- Flood-risk model — D8 flow accumulation

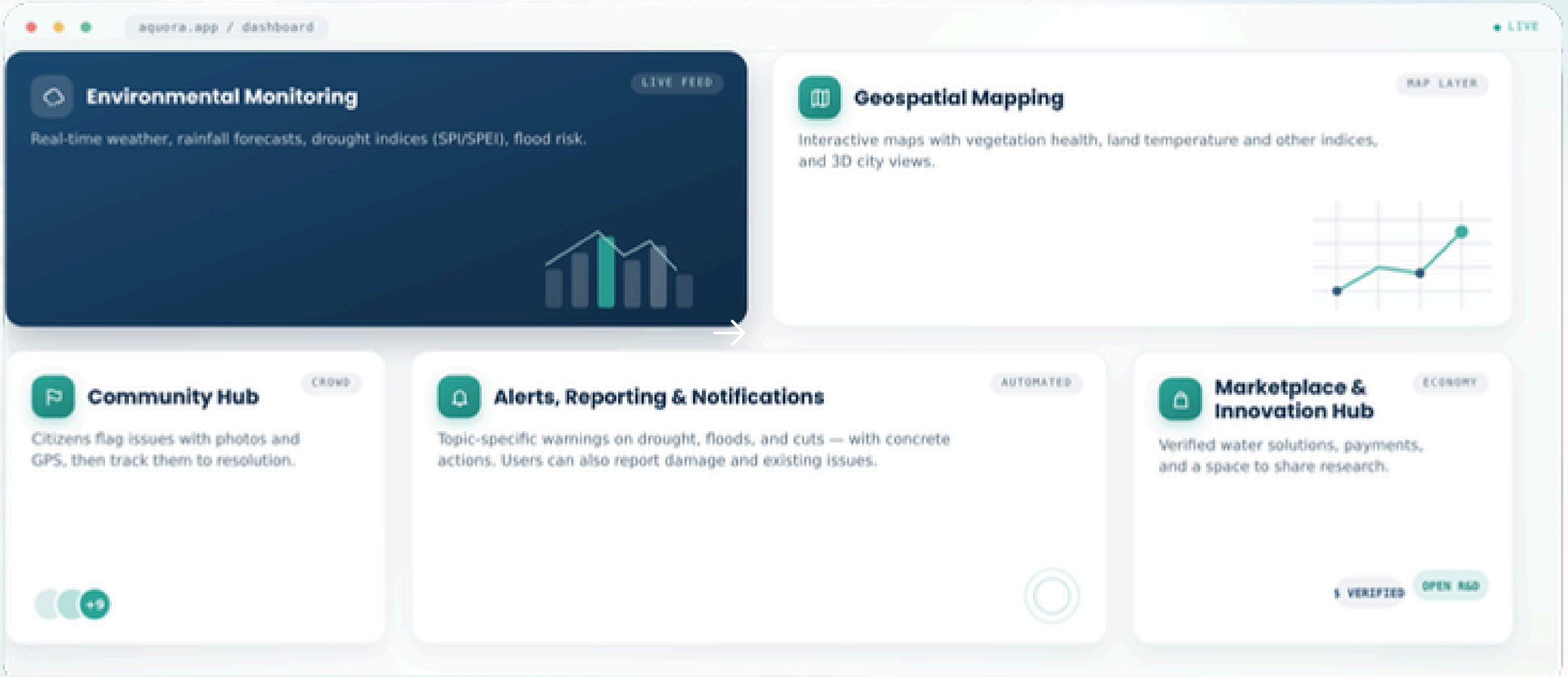
NDVI / NDWI band math on live satellite tiles

Platform Infrastructure

- SQLite — users, alerts, community, marketplace
- DNS MX + SMTP — auth & notification delivery
- Seeded demo data — cities, catalogues, samples

HOW IT WORKS

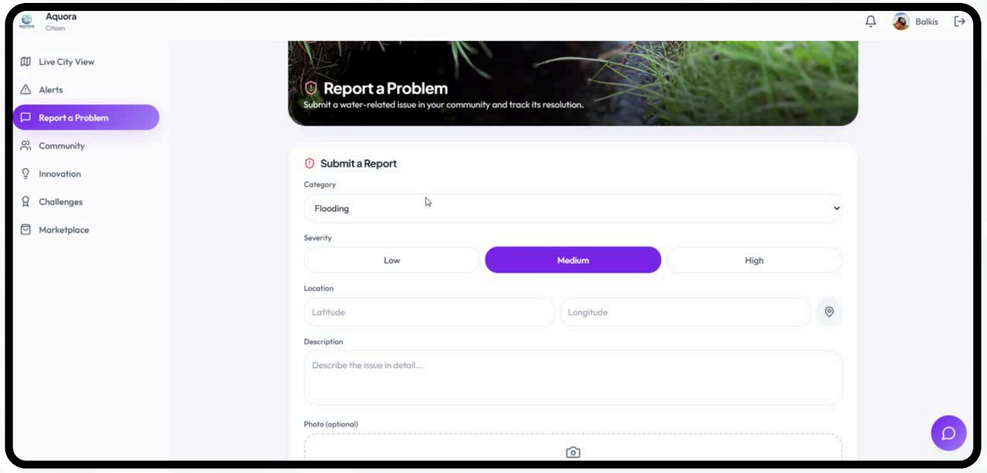
Five functions, one platform



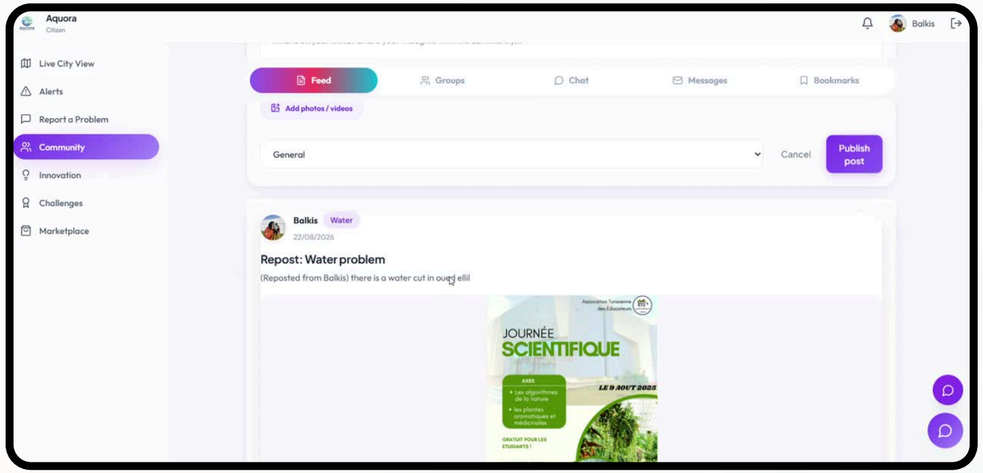
What a citizen can do?



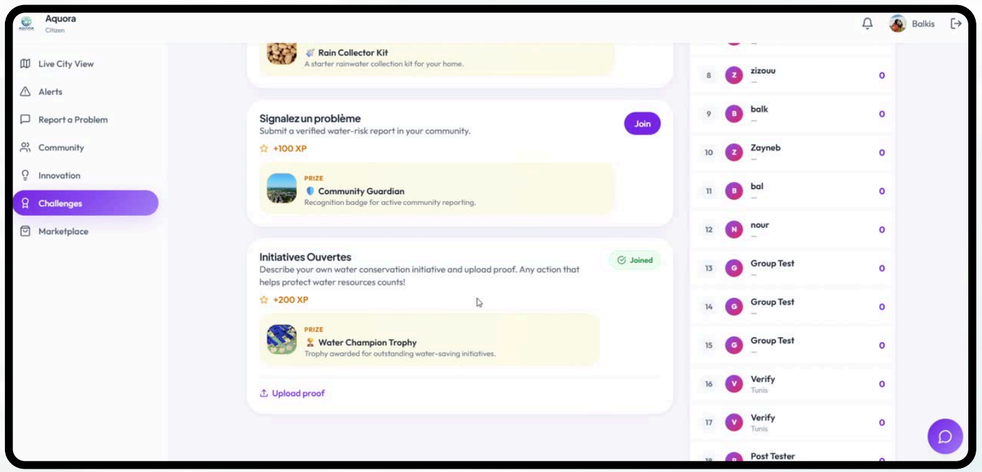
Access to information: See the city in 3D mode with climate information



Integration: Contribute to field truth and mitigation action through reporting tool

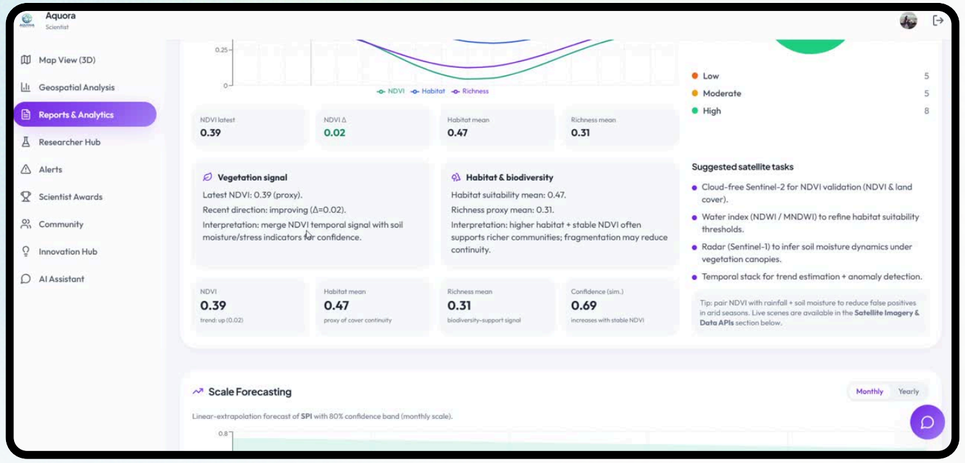


Networking: Connect with other citizens, scientists, and businesses

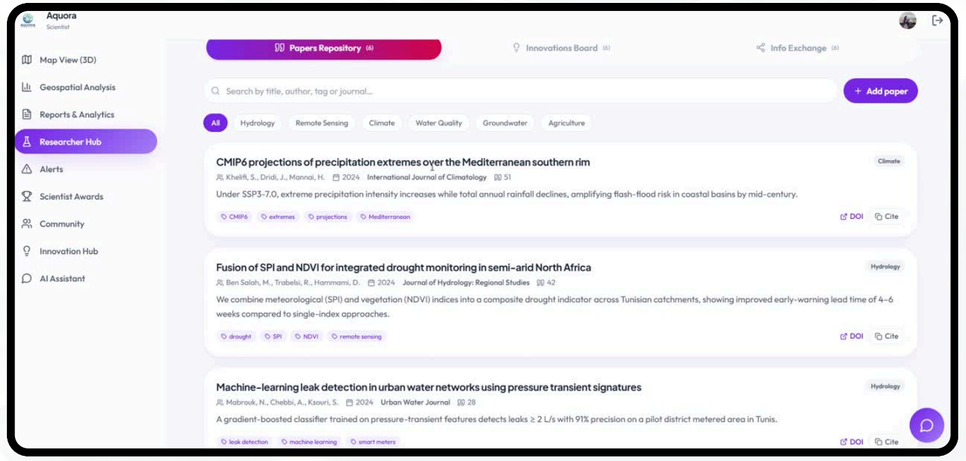


Taking initiatives: Through local actions, innovations, or open competitions

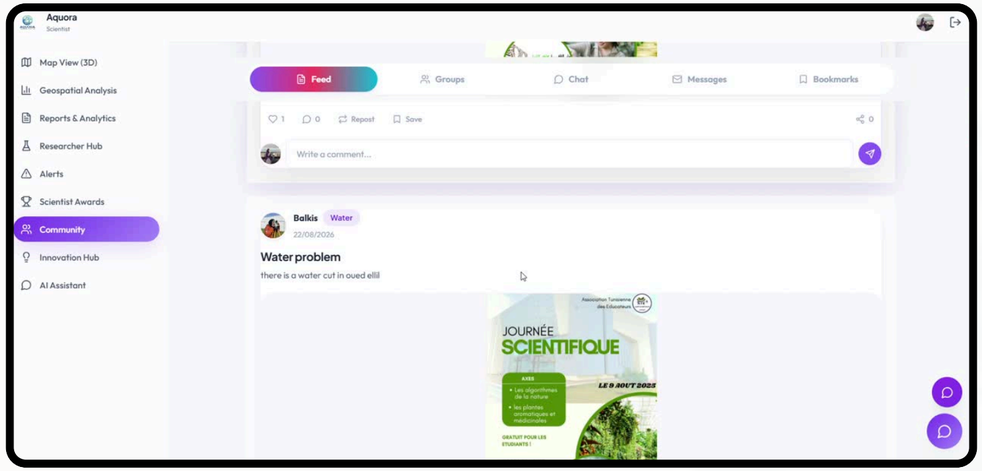
what can a Scientist do?



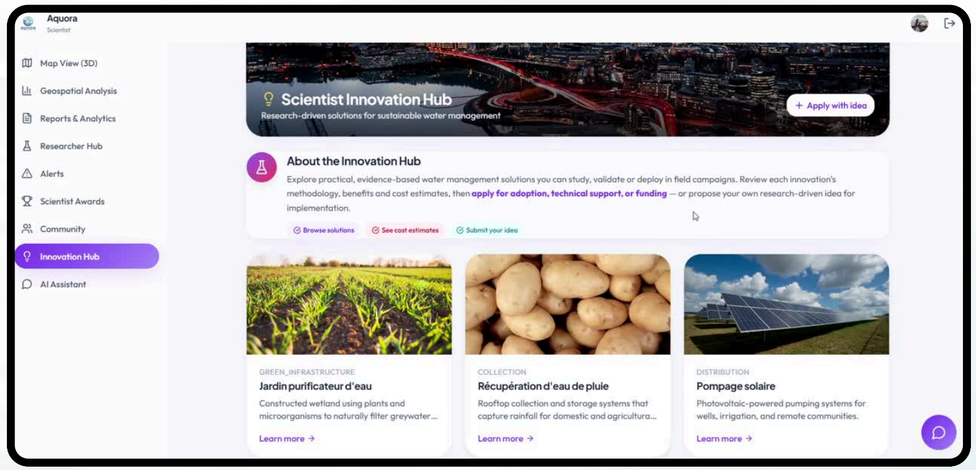
Access to information: time series analysis, satellite images, stats....



Integration: integration wth a community of researchers to post papers, data, innovations

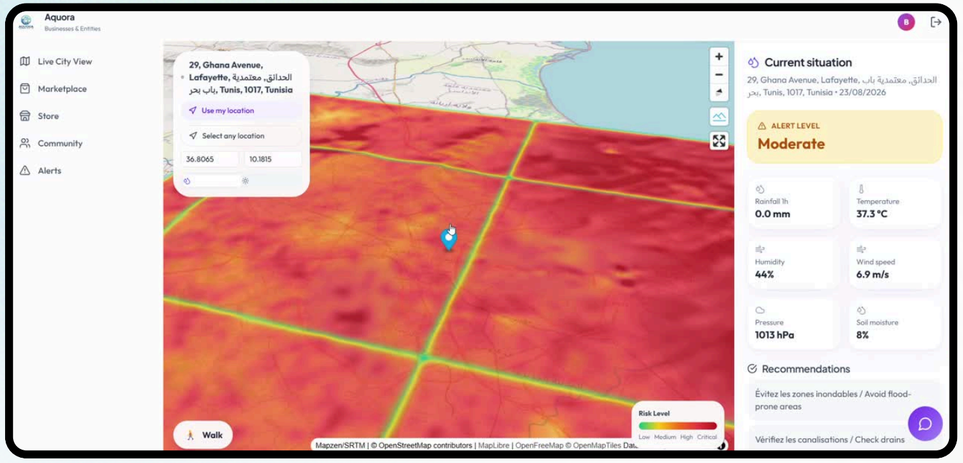


Networking: Connect with other citizens, scientists, and businesses

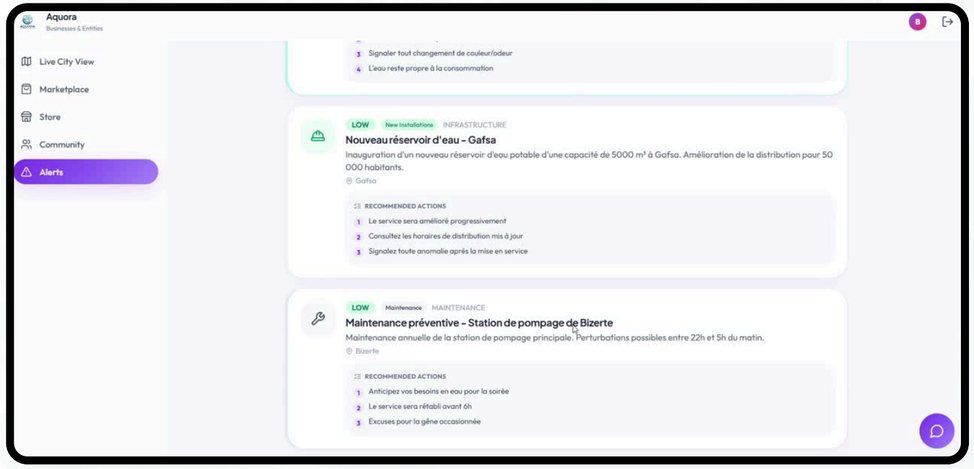


Taking initiatives: Through applying with innovation ideas

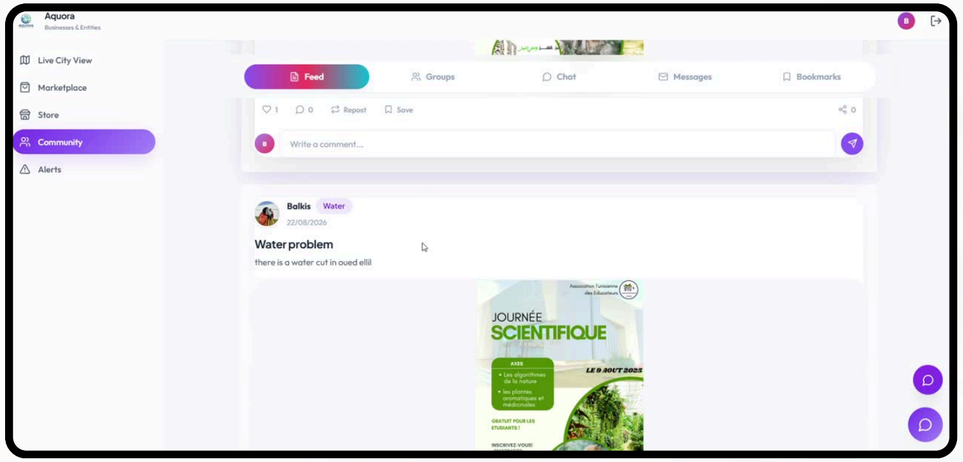
What can a Businesses do?



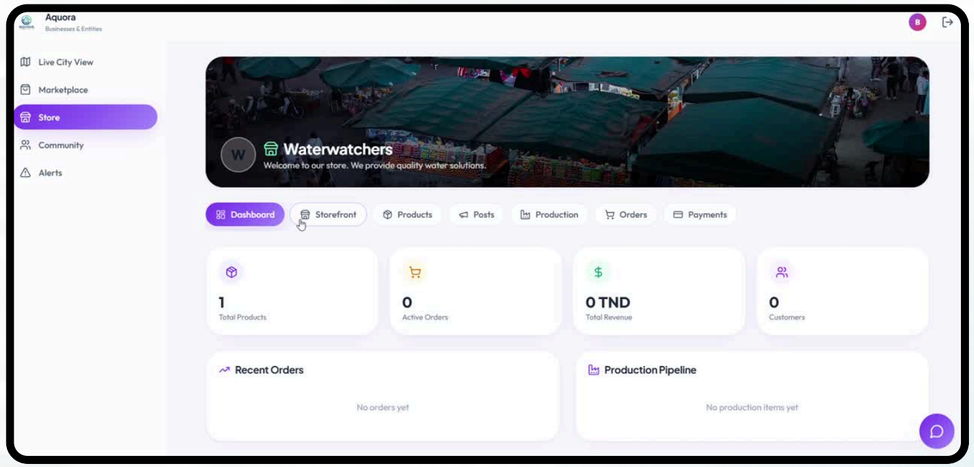
Access to information: See the city in 3D mode with customizedclimate information



Alerts and warning: to avoid damage and protect businesses



Networking: Connect with other citizens, scientists, and businesses



Taking initiatives: Through valuable products and services


Burst pipe on
Main Street
Photo attached
Municipality notified automatically
Track your report
Reported
In review
In progress
Resolved
Resolved
Thank you for helping keep our city better!
Researcher · Analyzing land degradation
Land Degradation Analysis
Low
Moderate
High
Very High
Trends
Hello, Sarah
How can we help today?
Report an issue
AI Assistant
Alerts
Marketplace
Maps & Data
Learn
Recent activity
Leak on Main Street
Resolved
Drought alert
Severe
Soil degradation
In review
Severe Drought Alert
Central Tunisia
AI Assistant
We recommend drip irrigation to save water and protect your yield.
Marketplace
Drip Irrigation Kit
Subsidized
View offer
Result
Yield
Stable
Water Use
-40%
Citizen
Farmer
Agency
Researcher
One platform,
Many roles,
Better outcomes.
Agency · Planning emergency response
Risk Zones
Low
Moderate
High
Very High
Resources
Water Trucks
Shelters
Medical Units

BUILT TO SCALE

Designed to handle growth & high demand

Cloud Infrastructure – Coming Soon

We will deploy on the cloud (**AWS / Azure / GCP**) to support more users and ensure flexible, reliable access.





Horizontal Scalability

As traffic increases, additional service instances can be added to handle more users simultaneously.

Containerized & Orchestrated

Docker and Kubernetes will allow services to scale independently based on demand.





Scalable Data Layer

Our database and storage will be designed to handle growing volumes of geospatial data.

Our goal : A platform that grows with its users, from local communities to national and regional scale.

BUILT TO BE MAINTAINABLE

Reliable today, ready for tomorrow

Modular Architecture

Components can be improved or replaced independently, reducing the impact of future changes.



Easier updates, reduced risk

Caching & Optimization – Planned

Caching & database optimization will help maintain fast response times as usage increases.



Better performance under heavy load

Clean, Maintainable Code

Coding standards and documentation will make the platform easier to understand, test, & extend.



Faster fixes and troubleshooting

Security by Design

Security practices will be integrated from the beginning to protect data and users.



Your data stays secure and trusted

Our Goal: To continuously evolve our platform, enhancing its performance & user experience to serve more users, better, every day.

What we need to scale ?

Three phases, one clear path

Phase	Ask	Unlocks
Pilot — 6 mo	$25K–$40K	1 municipality live, 100 active users, data validated with a local water authority + university
Regional — 6 mo	$50K–$80K	3 municipalities, 1,000 users, marketplace + innovation hub launched
National — 12 mo	$100K–$150K	Mobile apps, national weather/hydro integration, 25,000 users

Total: $175K–$270K over 24 months

> We’re not proposing just a concept , we’re proposing a real, actionable rollout

Tools & Tech Stack Used

FRONTEND

- **React** — UI Library
- **Vue.js** — Progressive JavaScript Framework
- **React Native** — Mobile App Development

3D VISUALIZATION

- **Three.js** — 3D Graphics Library
- **CesiumJS** — 3D Globe & Map Visualization

SEARCH

- **Elasticsearch** — Search & Analytics Engine

BACKEND

- **Node.js** — JavaScript Runtime
- **Python** — Programming Language
- **FastAPI** — Modern Web Framework

REAL-TIME

- **Socket.IO** — Real-time Bidirectional Communication
- **WebSocket** — Full-duplex Communication

DATABASE & DEVELOPMENT

- **PostgreSQL** — Open-source Relational Database
- **VS Code** — Lightweight & Powerful Code Editor

GIS & MAPPING

- **Mapbox** — Maps & Location Platform
- **Leaflet** — JavaScript Library for Interactive Maps

BACKEND CAPABILITIES

- **Weather** — Open-Meteo API
- **Hydrology** — Flood Risk Modeling
- **Drought** — SPI & Drought Analysis
- **Forecast** — Rainfall Prediction
- **OSM** — OpenStreetMap Integration

TECH STACK SUMMARY

React	Vue.js	React Native	Node.js	Python	FastAPI	Three.js	CesiumJS	Socket.IO	WebSocket	PostgreSQL	VS Code	Elasticsearch
UI Library	Framework	Mobile	Runtime	Language	Web Framework	3D Library	3D Globe	Real-time	Protocol	Database	IDE	Search Engine

PROJECT: AQUORA v2.0
National-scale water resilience digital ecosystem for Tunisia

PROJECT ROOT
c:\balkis\professional life\HydroTwin\HydroTwin_Community\

DATABASE
SQLite (current)
PostgreSQL / PostGIS (target)

AUTHENTICATION
JWT (python-jose) + bcrypt
Secure & Scalable

Our Vision

A collaborative space where data becomes action — helping communities build water resilience and food security across Africa, one region at a time.

THANK YOU

FOR YOU ATTENTION

Team AGM POWER · Balkis Selmi · Zayneb Ben Rajab · Nour Elimen Sbaiti
Hackathon SpatiELLES 2026